

When Meaning Gets Lost in Translation: Evaluating Semantic Alignment of LLM-Generated Dublin Core Metadata

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Abstract

"Can AI preserve the cultural meaning that human catalogers carefully construct?"

Cultural heritage metadata encodes community values and memory. This study investigates the semantic gap between LLM-generated and expert-curated Dublin Core records, exploring how AI systems risk imposing dominant cultural frameworks on materials requiring nuanced description.

Research Methodology

This study employs a comparative evaluation design using real-world Korean museum datasets.

Phase 1: Dataset Construction

Records drawn from the Seoul Museum of History featuring high cultural sensitivity (e.g., Joseon Dynasty artefacts, Prehistoric tools). These items require domain knowledge outside the primary training distributions of general-purpose LLMs.

Phase 2: LLM Evaluation

Testing LLMs across three prompting strategies:

- Zero-shot:** Baseline generation
- Few-shot:** Supplying expert-curated exemplars.
- Chain-of-Thought (CoT):** Forcing explicit reasoning prior to generation.



Scan for Details

Keywords

#DublinCore #LLMs
#CulturalHeritage
#SemanticAlignment
#MeaningDrivenAI

Research Flow

1. Dataset Construction

2. LLM Metadata Generation

3. Quantitative Evaluation (LaBSE)

4. Qualitative Evaluation (Expert Panel)

Concept: Semantic Flattening

The unintended replacement of culturally specific terminology with a generalized equivalent.

Rich Cultural Meaning
"Sanggam inlay technique"

Pressure of Dominant Defaults

Generic Equivalent
"Patterned design"

Illustrative Example: The Celadon Vase

How does Semantic Flattening manifest in practice? Consider the official record for the Celadon Prunus Vase with Inlaid Cloud and Crane Design.



EXPERT-CURATED BASELINE (Preserved Meaning)	ZERO-SHOT LLM OUTPUT (Semantic Flattening)
- Celadon (청자) - Goryeo dynasty (12th C) - Maebyeong (prunus vase) - Sanggam inlay technique	- Green ceramic - Believed to be East Asian - Vase [Omitted entirely]

Diagnostic Result: The unique historical context, object-type term, and technique-specific term are completely flattened into dominant-culture defaults.

DCMI 2026 & Future Impact

By measuring **Semantic Fidelity** (via multilingual embeddings like LaBSE) and **Cultural Specificity** (via expert panel reviews), this research provides a replicable framework for diagnosing meaning loss.

Mitigating Semantic Flattening directly answers the DCMI 2026 call for **Meaning-Driven AI**: using metadata to align systems with human values, rather than allowing algorithms to override them.